Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/29/2011
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	Comparative Analysis of Charged Particle-Induced Autosomal Mutations in Murine Tissues and Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	04/01/2007	**End Date:**	03/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Project extended until 3/31/2012 per C. Guidry/JSC (Ed., 2/14/2011)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Turker, Mitchell (Oregon Health & Science University) Grygoryev, Dmytro (Oregon Health & Science University) Dan, Cristian (Oregon Health & Science University) Gauny, Stacey (Lawrence Berkeley National Laboratory) Grossi, Gianfranco (Universita Federico II)		
Grant/Contract No.:	NNJ07HC72I		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Mutations are required in the transition from normal cells to cancer cells. Solid cancers in people arise primarily in epithelial cells. This proposal addresses the carcinogenic risk of exposure to energetic protons, the most abundant ion in galactic cosmic radiation (GCR) and solar particle events. The toxic effects of these exposures will also be measured. The studies outlined quantify mutation induction at a model, autosomal genetic locus (Aprt) located on mouse chromosome 8. C57BL/6 x DBA2 F1 Aprt heterozygous mice or kidney epithelial cells from this mouse strain will be irradiated to determine dose responses for cell killing and Aprt mutation in kidney epithelial cells exposed in the mice, or in kidney epithelial cells irradiated in culture. This will allow assessment of the role of the tissue microenvironment on modulation of mutation risk and on persistence of radiation-induced mutants in the kidney. Molecular analysis will determine the mechanisms leading to proton-induced Aprt mutants in exposed kidneys or in kidney cells irradiated in culture. It is predicted that proton exposure will result in mutants that lose the Aprt gene and other genes that are located in nearby regions of mouse chromosome 8. Studies are proposed to test if protracted exposure lessens the frequency of proton-induced mutants while also lowering the risk of killing kidney epithelial cells. The toxic and mutagenic effects of sequential exposure to protons and iron ions will be assessed (with iron ions serving as a representative heavy ion in the GCR) to test whether the risks are additive. Results from these studies will help reduce uncertainties in estimating the risk of cancer for astronauts exposed to protons alone or in mixed radiation fields. They will also provide detailed knowledge of the types of mutations that are created by these exposures that can contribute to the development of cancer.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	This research will have a direct impact on our understanding of proton-induced toxicity and mutations in a mammalian epithelial tissue. This has relevance to cancer therapy involving proton exposures that not only target malignant tissue, but which also involve limited exposure of normal tissues that can develop late complications as a result of proton exposure. In addition, our in vivo/in vitro approach also illuminates the importance of the normal tissue environment on the modulation of radiation sequelae.		
Task Progress:	During the Spring 2010 run at BNL, studies were initiated to compare the effect of fractionated exposures to protons vs. acute exposures to protons in the kidney epithelial model system. The results to date indicate sparing for toxicity when the dose is fractionated. Fractionation also was sparing for mutant frequency. In the Fall 2010 run at BNL, studies were performed with sequential irradiations of cultured mouse kidney epithelial cells exposed to 1 GeV/amu Fe ions and 1 GeV protons. Doses were separated by ~24 hours. Both exposure regimens were considered: protons followed by Fe ions or Fe ions followed by protons. The results for these sequential irradiations were benchmarked against graded, acute exposure to each ion individually. The results suggest that sequential exposure to these key ions are additive. Studies continued for B6D2F1 mouse kidney epithelial cells following acute exposures to graded doses of 1 GeV protons in situ. Proton exposure elicits a mild toxicity as a function of dose at short times post-exposure, however recovery was observed across all doses with increasing time. Analysis of the final data set for the corresponding Aprt mutant frequency results is ongoing. A large series of proton-induced mutants have been collected from kidney epithelial cells established after in situ exposures harvested at early and late harvest times. The results indicate that the mutant spectrum for proton-irradiated kidney epithelium contains a modest increase in the proportion of Aprt mutants arising via mitotic recombination in the proton-irradiated group when contrasted against Aprt mutants arising spontaneously A comparison of proton-induced mutants with those arising after exposure to a lower dose of 1 GeV/amu Fe ions indicates that there is a marked effect of LET on the mutant spectra. This was true whether cells were exposed in vitro or in situ.		

Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Kronenberg A, Dan C, Sandfort K, Turker M. "Evidence for genomic instability in normal mouse epithelium exposed to charged particle radiation in situ." Presented at the NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010. NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010. http://www.dsls.usra.edu/meetings/hrp2010/pdf/Radiation/1132Kronenberg.pdf , Feb-2010
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Sudo H, Wiese C, Dan C, Turker M. "Comparative biology approaches for charged particle exposures and cancer development processes." Presented at the Committee on Space Research (COSPAR) 2010 38th Scientific Assembly, Bremen, Germany, July 18-25, 2010. COSPAR Abstract Book. Committee on Space Research (COSPAR) 2010 38th Scientific Assembly, Bremen, Germany, July 18-25, 2010. https://www.cospar-assembly.org/abstractcd/COSPAR-10/ ; go to Commission F to search for abstract , Jul-2010
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Dan C, Turker M. "Autosomal mutations in murine kidney epithelial cells exposed to energetic protons." Presented at the 56th Annual Meeting of Radiation Research Society, Maui, HI, September 25-29, 2010. 56th Annual Meeting, Radiation Research Society, September 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=9e011f35-da29-44e9-81ea-4b52cf617fb4&cKey=ccfff2b9-03df-4e18-b3c1-20acc897b96a&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Articles in Peer-reviewed Journals	Dan C, Grygoryev D, Sandfort K, Connolly M, Cross B, Lasarev M, Kronenberg A, Turker MS. "Marked aneuploidy and loss of multiple chromosomes are common in autosomal mutants isolated from normal mouse kidney epithelium." Genes, Chromosomes and Cancer, 2011 Apr;50(4):239-49. Epub 2011 Jan 19. PMID: 21254298 ; http://dx.doi.org/10.1002/gcc.20849 , Apr-2011